

3D Coverage Performance of VCSEL Arrays in Indoor OWC Systems for 6G Applications

Hossein Safi, Mohammad Taghi Dabiri, Sina Babadi,
Iman Tavakkolnia, *Member, IEEE*, and Harald Haas, *Fellow, IEEE*

Abstract—To meet key performance indicators for future telecom networks, such as ultra-high data rates, energy efficiency, and flexible coverage, this paper investigates the potential of vertical-cavity surface-emitting laser (VCSEL)-based indoor optical wireless communication (OWC) systems in dynamic communication environments. We analyze the influence of system parameters, including optimal coverage height, divergence angle, and the allocation of VCSEL beams to users, on total coverage performance and system efficiency. Our findings demonstrate that optimizing the system for enhanced 3D coverage at higher heights improves overall coverage performance but reduces network capacity due to limited availability of VCSEL beams at lower heights. Additionally, by prioritizing coverage over spatial multiplexing and allocating multiple laser links to individual users, we achieve 100% coverage while maintaining a balance between coverage and capacity. These results highlight the critical role of dynamic beam allocation and divergence angle optimization in enabling comprehensive 3D coverage, paving the way for efficient and high-performance VCSEL-based OWC systems in 6G and beyond.

Index Terms—Divergence angle, Optical wireless communications (OWC), vertical cavity surface emitting laser (VCSEL), 3D Coverage, 6G.

I. INTRODUCTION

The rapid evolution of communication technologies is driven by the increasing demand for seamless connectivity and the transformative potential of data-intensive next-generation applications [1]. To address the unprecedented connectivity requirements of 6G and beyond, fully integrated networks of networks (NoN) have emerged as a pivotal solution [2]. By seamlessly integrating diverse networks, 6G aims to achieve ultra-low latency, massive connectivity, and high energy efficiency. As an important component of fully integrated NoN, optical wireless communication (OWC) has emerged as a promising solution to address the connectivity demands of 6G and beyond [3], [4]. Exploiting its vast unregulated spectrum and the capability to achieve data rates exceeding 100 Gbps [5], OWC is ideally suited for bandwidth-intensive applications such as high-definition video streaming, augmented reality, and virtual reality. Moreover, the confined nature of OWC enhances data security, a critical feature for environments like hospitals, military installations, and financial institutions. By integrating seamlessly with other access technologies, OWC not only complements existing networks but also enhances

connectivity and performance, particularly in RF-constrained scenarios or where additional bandwidth is essential [6], [7].

Recent advancements in OWC have highlighted the potential of vertical-cavity surface-emitting lasers (VCSELs) to achieve data transmission speeds of tens of Gigabits per second (Gb/s) while adhering to stringent eye-safety regulations [8]. Leveraging the inherent directionality of light, VCSELs facilitate the development of optical access points (APs) capable of delivering aggregate data rates exceeding 1000 Gb/s, and make it feasible to meet the key performance indicators of 6G [3], [8]. VCSELs stand out as a preferred choice over LEDs and other laser diodes due to their numerous advantages, including high power efficiency, extensive modulation bandwidth, and precise beam control. Unlike edge-emitting lasers, VCSELs are cost-effective, readily fabricated, and can be arranged into compact two-dimensional arrays. Additionally, their enhanced reliability, low manufacturing costs, and circularly symmetric beam profile simplify system integration and alignment [9].

Despite its advantages in indoor environments, OWC faces challenges such as link blockages caused by moving objects like people or furniture. These obstacles can disrupt the line-of-sight path between the transmitter and receiver, resulting in signal loss and degraded performance. To address this challenge, VCSEL-based OWC systems utilize structured grids of VCSEL arrays to form an array of optical beams [3], [10]–[13]. These lenses function to direct the beams in predefined but fixed directions, ensuring coverage over the target area. This configuration enhances link robustness and maintains connectivity even when individual beams are blocked [12].

Prior art in this context, e.g., [10]–[13], primarily focused on ensuring single-beam access for each user in practical VCSEL array-based OWC systems. To achieve this, the AP architecture typically employed multiple transmitter elements, each oriented to cover a specific area within the room. Each transmitter element comprises a VCSEL array in which each VCSEL in the array is individually addressable and a plano-convex lens with a fixed focal length. While designed to minimize interference and provide full coverage, this approach has some inherent limitations.

First, such APs are designed to provide coverage at a fixed height. Specifically, the system was designed and optimized to minimize interference (by avoiding the overlap of adjacent beams on the receiver plane) and provide coverage on the receiver plane, which is located at a certain height. This assumes that the receiver does not make vertical movement, which is not a practical assumption, especially if the user is communicating with a handheld device.

H. Safi, I. Tavakkolnia, S. babadi, and H. Haas are with the Electrical Division, Department of Engineering, University of Cambridge, Cambridge, UK, e-mails: {hs905, it360, sb2770, huh21}@cam.ac.uk.

M. T. Dabiri is with the Department of Electrical Engineering, Qatar University, Doha, Qatar (e-mail: m.dabiri@qu.edu.qa).

Second, the number of VCSELs in the array is chosen to guarantee coverage at a predefined height. However, this is not energy-efficient because the received optical beamwidth varies with height (it widens as the distance from the AP increases). Therefore, the number of required VCSELs in the array for providing full coverage differs at different heights. This implies that utilizing all VCSELs at lower heights may be unnecessary, potentially leading to energy waste.

Third, the previous AP structures assumed that the divergence angle of the output beam from the transmitter is fixed, which means that the beamwidth at the receiver plane remains constant. In practice, a fixed divergence angle leads to inefficient coverage and potential misalignment, particularly in dynamic environments where users may be moving or the geometry of the communication link changes. This highlights the need for more flexible beam steering mechanisms and dynamic divergence angle adjustments to ensure reliable and efficient performance under varying conditions.

Motivated by the limitations of existing systems, this paper proposes a VCSEL-based AP architecture designed to provide reliable coverage across varying heights of the receiver plane (i.e., 3D coverage) in indoor scenarios. More specifically, we introduce the use of variable focus lenses in such AP architecture, which enable adaptive divergence angle adjustments. Variable focus lenses have emerged as a promising solution for dynamic beam control in OWC systems [14], [15]. These lenses can adjust their focal length in response to an applied voltage, offering a compact and efficient method to modify the divergence angle of output optical beams in confined spaces [14]. The ability to electrically tune the beam divergence not only enhances system flexibility but also improves coverage, reduces interference, and increases data rates in dynamic indoor scenarios.

We further conduct extensive simulation to analyse link coverage at various heights within the considered communication cell. Additionally, we study the effect of varying divergence angles of the output optical beams on the AP's link coverage and aim to identify the optimal divergence angle for maximizing coverage percentage under different scenarios. Our results highlight that determining the optimal values for the divergence angle requires balancing a trade-off between broader beam coverage and the interference of adjacent beams within the cell. Our study provides a benchmark for optimizing the divergence angle of VCSEL arrays to ensure reliable link coverage. These findings can serve as a foundation for future studies to refine beamforming strategies and enhance the performance of VCSEL-based OWC systems in dynamic indoor environments, thereby supporting the development of sustainable and resilient 6G networks.

The remainder of this paper is organized as follows. In Section II, we present the system model of the considered VCSEL-based indoor OWC system. Simulation results and discussion of the link performance under different scenarios for providing coverage are presented in Section III. Finally, Section IV concludes the paper with key findings and outlines potential directions for future research.

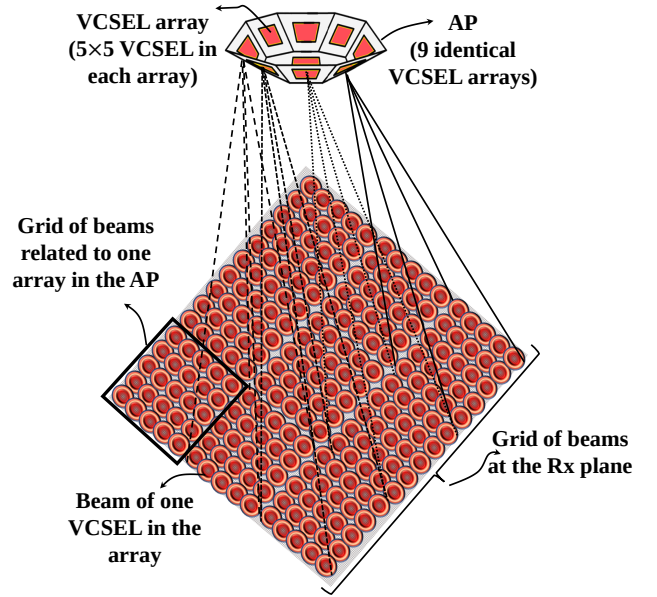


Fig. 1. Indoor OWC network employing a dense array of VCSELs for complete beam coverage within the room.

II. SYSTEM MODEL

This section provides an overview of the proposed indoor OWC system, which utilizes a grid of beams using VCSEL arrays. We begin by describing the system architecture and outlining its operational principles. Next, we discuss the characteristics of the optical beams used in the system and analyze their geometric configuration. Finally, we establish a theoretical framework for evaluating system performance, with a focus on the signal-to-interference-plus-noise ratio (SINR) as a measure of coverage.

A. Geometric Configuration

Fig. 1 shows an indoor OWC network utilizing a dense array of VCSELs for complete beam coverage. The proposed architecture consists of nine identical transmitter elements, each incorporating a 5x5 VCSEL array, resulting in a total of 225 VCSELs (optical beams) that are operating at wavelength λ . As previously mentioned, this design ensures that each VCSEL array effectively covers a specific region within the indoor cell, thereby providing ubiquitous and efficient coverage in the receiver plane, as depicted in Fig. 1.

Due to the need for high packing density, efficient coupling to multimode fibers without external optics, robust thermal and electrical management, and enabling functionalities like redundancy by allowing spare VCSELs to be closely integrated with active ones within the same pixel, the pitch distance between individual VCSELs is minimal [16], [17]. This can lead to potential beam overlap, or interference, at the receiver plane, particularly when multiple beams are emitted from closely spaced sources. To mitigate this issue, optical components, such as lenses, are placed in front of each VCSEL. These lenses function as beam-steering elements, refracting the emitted light beams in different directions to enhance

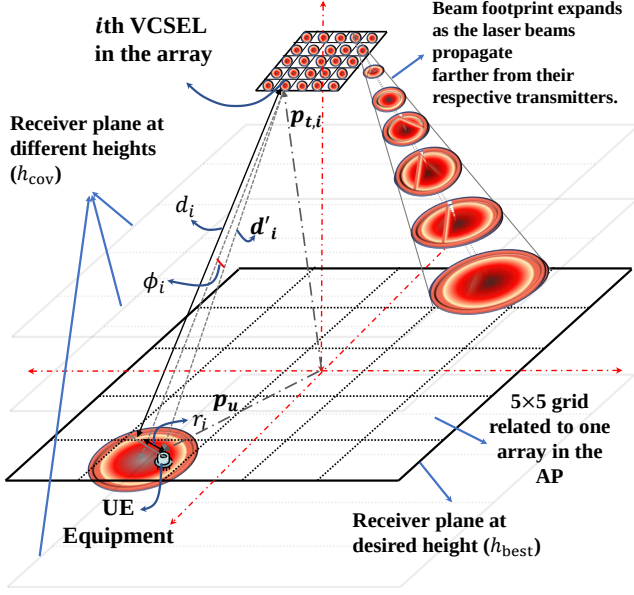


Fig. 2. Downlink Communication Geometry for a VCSEL Array in the AP.

spatial resolution. Additionally, the received beamwidth at the receiver plane is influenced by the beam's propagation path length and the divergence angle of the output beam from the transmitter. The divergence angle is a crucial parameter that must be carefully optimized during the access point design to achieve the desired spatial resolution and ensure consistent coverage across varying receiver plane heights.

B. Beam Characteristics

The output beam profile of a VCSEL can be approximated as Gaussian, especially at lower bias currents, as stated in [13]. Under these conditions, mode competition—where multiple optical modes within the laser cavity compete for gain—is minimal, resulting in a stable and well-defined Gaussian beam at the output of the optical source. Consequently, considering a Gaussian beam at the output of the VCSELs, the normalized spatial intensity distribution of the i th beam on the receiver plane is given by [18]

$$I_i(r, d) = \frac{2}{\pi\omega^2(d_i)} \exp\left(-\frac{2r_i^2}{\omega^2(d_i)}\right), \quad (1)$$

where r_i is the radial distance from the i th beam center. Furthermore in (1), $\omega(d_i)$ denotes the beamwidth as a function of link distance d_i , and it can be approximated as

$$\omega(d_i) \simeq \omega_{o,i} \sqrt{1 + \left(\frac{\lambda d_i}{\pi\omega_{o,i}^2}\right)^2}, \quad (2)$$

where λ denotes the VCSELs' operating wavelength. Moreover, in (2), the beam waist for the i th beam, i.e., the point where the output beam diameter is at its minimum, is defined as $\omega_{o,i} = \frac{\lambda}{\pi\theta_{\text{div},i}}$, where $\theta_{\text{div},i}$ is the divergence angle of the output optical beam from the i th VCSEL. In this work, we define the divergence angle based on the full width at half

maximum criterion, where the power intensity drops to 50% of its peak value.

From Fig. 2, the distance vector from the i th VCSEL in the array to the user is expressed as $\mathbf{d}'_i = \mathbf{p}_{t,i} - \mathbf{p}_u$, where $\mathbf{p}_{t,i}$ is the position vector of the i th VCSEL and $\mathbf{p}_u$ is the user position vector. Consequently, the corresponding Euclidean distance can be calculated as $d'_i = \|\mathbf{d}'_i\|$, where $\|\cdot\|$ is the Euclidean norm. Let ϕ_i represent the radiance angle of the i th VCSEL beam. From this, one can obtain $r_i = d'_i \sin \phi_i$, and $d_i = d'_i \cos \phi_i$. Substituting these expressions into the intensity equation (1), the revised formulation becomes

$$I_i(\phi_i, d'_i) = \frac{2}{\pi\omega^2(d'_i \cos \phi_i)} \exp\left(-\frac{2(d'_i \sin \phi_i)^2}{\omega^2(d'_i \cos \phi_i)}\right). \quad (3)$$

C. SINR Framework

The received optical power at the receiver detector plane can be determined by integrating the received intensity over the effective area of the detector [13]. Therefore, the downlink channel gain between the i th VCSEL and the user can be expressed as

$$h_i = I_i(\phi_i, d'_i) A_D G_D \cos \psi_i \Pi\left(\frac{\psi_i}{\Psi_D}\right), \quad (4)$$

where A_D represents the effective area of the detector, G_D denotes the detector gain, and ψ_i represents the incidence angle of the beam with respect to the normal vector of the receiver. Furthermore in (4), Ψ_D is the receiver field-of-view and $\Pi(\cdot)$ denotes the rectangular function which is defined as

$$\Pi\left(\frac{\psi_i}{\Psi_D}\right) = \begin{cases} 1, & \text{if } \psi_i \leq \Psi_D, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Finally, assuming the i th VCSEL as the desired optical transmitter, the downlink SINR for the proposed setup is given by

$$\text{SINR} = \frac{P_{t,i} h_i}{\sum_{\substack{j=1 \\ j \neq i}}^{225} P_{t,j} h_j + \sigma_n^2}, \quad (6)$$

where $P_{t,i}$ is the transmit optical power of the i th VCSEL and σ_n^2 is the total noise power at the receiver.

III. SIMULATION AND DISCUSSION

In this section, we provide the simulation results to evaluate the performance of the proposed VCSEL-based indoor OWC system, focusing on coverage and system efficiency under different configurations. We analyze the impact of varying system parameters, such as the optimal height, divergence angle, and the number of VCSEL beam allocated to the user, to understand their effects on the coverage, link capacity, and overall system performance at various heights.

For the simulation, we consider an indoor environment with dimensions of $L \times W \times H$, where L is the length (10 meters), W is the width (10 meters), and H is the height (3 meters). The AP is installed at the center of the ceiling, capable of generating 225 distinct laser beams to provide comprehensive

coverage in the indoor space. The power allocated to each VCSEL operating at 940 nm is $P_{t,i} = 5\text{mW}, \forall i$ which is aligned with eye-safety regulations [11]. We also assume all of the VCSEL in the array follow a same Gaussian beam profile [8], [11], [12].

As discussed, the defining feature of this topology, which sets it apart from conventional indoor optical systems with a single optical source, is the deployment of a dense array of VCSELs. Each VCSEL emits highly directional laser signals, with beam footprints expanding as the laser beams propagate farther from their respective transmitters. This characteristic leads to variations in coverage at different heights within the environment. We define h_{best} as the optimal height at which coverage and overall system performance are maximized.

To achieve coverage at h_{best} , the entire coverage area $L \times W$ is divided into 225 equal square subregions, with each VCSEL in the array precisely oriented to cover its designated subregion. The parameter $\theta_{\text{div},m}$ represents the maximum permissible divergence angle for the VCSEL beam corresponding to the shortest vertical link. This divergence angle, combined with the chosen value of h_{best} , plays a pivotal role in determining the coverage performance within the 3D space of the room, as examined in the subsequent sections.

A. Coverage Analysis at Different Heights

To obtain more insight into the overall system performance, we select $h_{\text{best}} = 1.5$ m, and begin by evaluating the coverage at different room heights¹. Accordingly, Fig. 3 illustrates the coverage for four different heights, i.e., $h_{\text{cov}} = 2, 1.5, 1$, and 0 , shown in Figs. 3a to 3d, respectively, where $\theta_{\text{div},m} = 6^\circ$. For better observation, we only show SINR over the threshold SINR (i.e., 5 dB) in the figures. As expected, the optimal coverage and performance are achieved at the height $h_{\text{best}} = 1.5$ m. At higher heights, the beam has not yet fully expanded, resulting in uncovered edges, which is undesirable, especially if the user makes vertical movements. In contrast, while lower heights do not face coverage gaps at the edges of the room, the whole coverage can be provided by fewer beams to increase efficiency and total capacity. For example, at a height of 1 meter, only 121 beams out of the total 225 beams can provide service, and this number decreases further to 49 beams at ground level.

To deeply understand the effect of choosing h_{best} on system performance, we provide the results for the coverage where $h_{\text{best}} = 1$ in Fig. 4. Comparing the results in Figs. 3 and 4, it is observed that as h_{best} decreases or approaches ground level, coverage and total capacity improve at lower heights, at the cost of losing coverage at higher heights. For instance, as seen in Figs. 3a and 4a, decreasing h_{best} to 1 results in more uncovered edges at a height of 2 m. However, at lower heights, the coverage improves, and more VCSEL beams remain usable. This trade-off emphasizes the importance of selecting an optimal h_{best} based on the expected user configuration and movement to balance performance across various heights. While lower heights may benefit from

more efficient beam usage, it is crucial to ensure that higher heights maintain sufficient coverage to support dynamic user mobility, particularly when the user is at varying heights. Furthermore, as observed in Figs. 3 and 4, uncovered spots remain in the horizontal plane of the receiver. To ensure comprehensive coverage and support user mobility within the horizontal plane, the appropriate beamwidth must be chosen based on the divergence angle.

B. Effect of Divergence Angle on Coverage

To investigate the effect of $\theta_{\text{div},m}$ on coverage, we maintain the setup of Fig. 3 but set $\theta_{\text{div},m} = 15^\circ$; and plot the results in Fig. 5. As clearly observed from this figure, changing the divergence angle significantly affects coverage. By comparing the results of Figs. 3 and 5, it is evident that a larger divergence angle leads to wider beam expansion, covering more marginal areas at higher heights. However, this increased divergence angle results in larger beam sizes at lower heights, which in turn exacerbates interference and degrades the SINR, ultimately decreasing the link capacity.

Thus, the optimal choice of $\theta_{\text{div},m}$ requires balancing a trade-off between a larger divergence angle for broader VCSEL beam coverage and a reduced divergence angle for better interference management. To this end, Fig. 6 shows the coverage as a function of $\theta_{\text{div},m}$ when $h_{\text{best}} = 1.5$ m, for four different values of h_{cov} . The results clearly highlight the importance of selecting an optimal $\theta_{\text{div},m}$ to maximize coverage percentage at different heights. The findings also demonstrate that as h_{cov} increases, i.e., when coverage is needed at higher heights, a larger divergence angle is required to cover more marginal areas. Another important observation is that for heights lower than h_{best} , the percentage of coverage remains almost constant. This might lead to the inference that optimizing the link for maximum coverage at higher heights, i.e., choosing a larger h_{best} , can provide better 3D coverage performance. However, although a larger h_{best} improves 3D coverage, the results in Figs. 4-5 indicate that increasing h_{best} reduces the number of VCSEL arrays capable of providing coverage at lower heights, ultimately decreasing the overall network capacity. Similarly, Fig. 7 illustrates the results from Fig. 6 for a new $h_{\text{best}} = 0.5$. Here, the findings reveal that reducing h_{best} also reduces the optimal value of $\theta_{\text{div},m}$ in order to maximize the coverage at the desired low height.

C. Multi-VCSEL Allocation for Improved Coverage

The results presented in Figs. 6 and 7 suggest that the current AP setup may not ensure 100% coverage at the desired height. However, in certain applications, particularly mobile scenarios, coverage may be prioritized over spatial multiplexing and the pursuit of higher capacity. To address this and enhance network resilience, multiple VCSELs within the array can be allocated to serve a single user. To investigate this approach, the results from Fig. 7 are re-analyzed in Fig. 8, where the target user is served by two VCSELs within the access point. The results clearly demonstrate that by sacrificing spatial multiplexing and allocating multiple laser links to each user (i.e., balancing the trade-off between coverage and capacity), 100% coverage can be achieved.

¹While we consider this as a practical height in our analysis, the results presented in this paper can be readily extended to any predefined height.

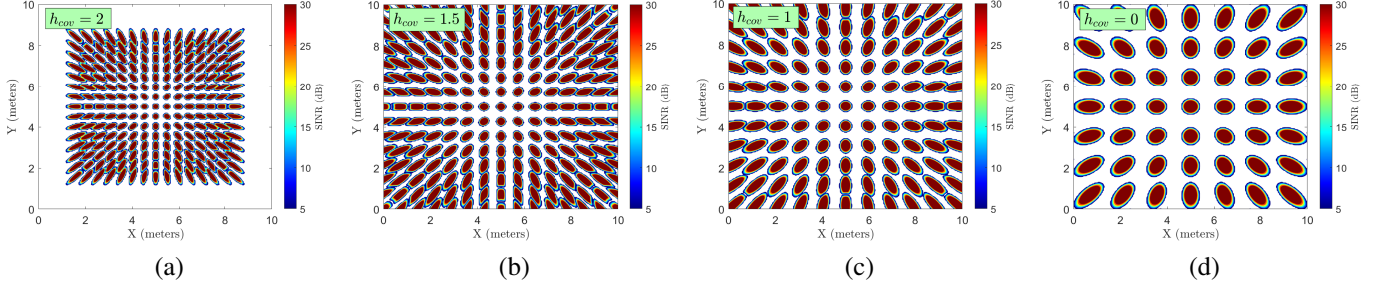


Fig. 3. Coverage performance for $\theta_{\text{div},m} = 6^\circ$ and $h_{\text{best}} = 1.5$ at different heights: (a) $h_{\text{cov}} = 2$ m, (b) $h_{\text{cov}} = 1.5$ m, (c) $h_{\text{cov}} = 1$ m, and (d) $h_{\text{cov}} = 0$ m.

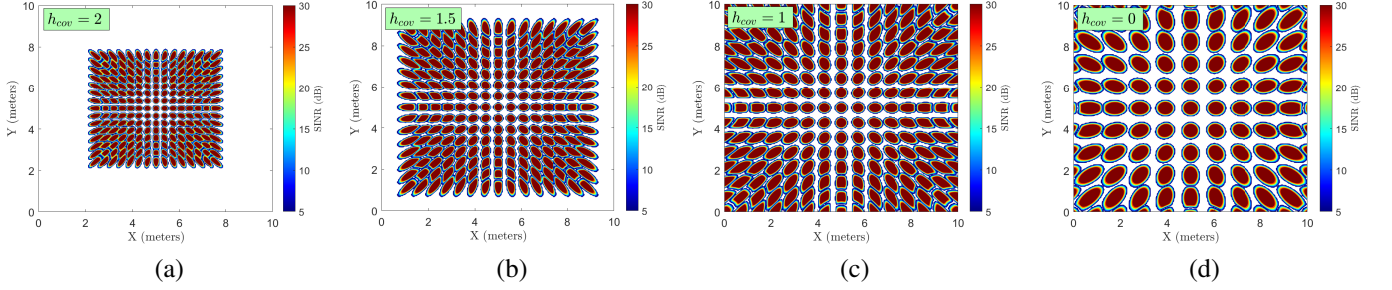


Fig. 4. Coverage performance for $\theta_{\text{div},m} = 6^\circ$ and $h_{\text{best}} = 1$ at different heights: (a) $h_{\text{cov}} = 2$ m, (b) $h_{\text{cov}} = 1.5$ m, (c) $h_{\text{cov}} = 1$ m, and (d) $h_{\text{cov}} = 0$ m.

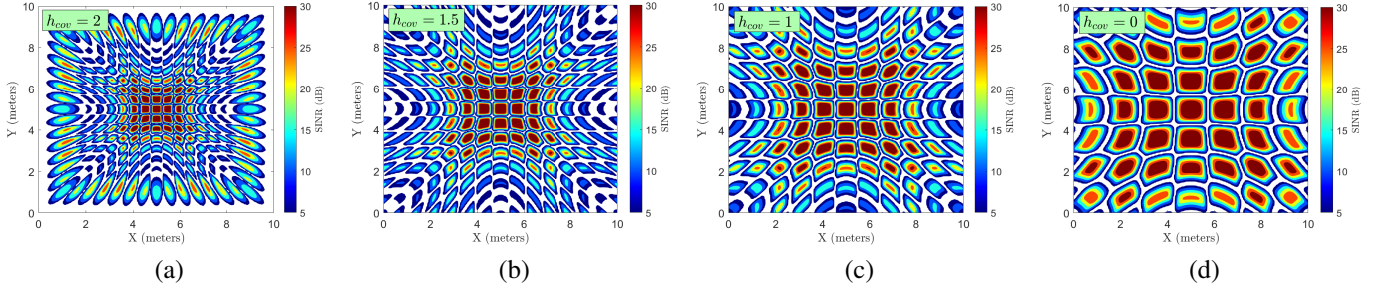


Fig. 5. Coverage performance for $\theta_{\text{div},m} = 15^\circ$ and $h_{\text{best}} = 1.5$ at different heights: (a) $h_{\text{cov}} = 2$ m, (b) $h_{\text{cov}} = 1.5$ m, (c) $h_{\text{cov}} = 1$ m, and (d) $h_{\text{cov}} = 0$ m.

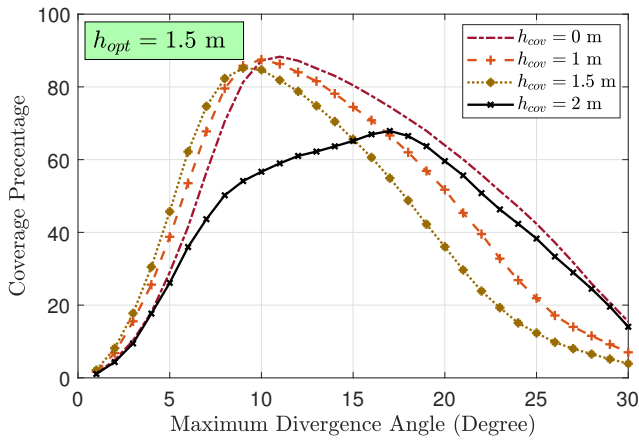


Fig. 6. Coverage as a function of $\theta_{\text{div},m}$ for $h_{\text{best}} = 1.5$ m and four different h_{cov} values.

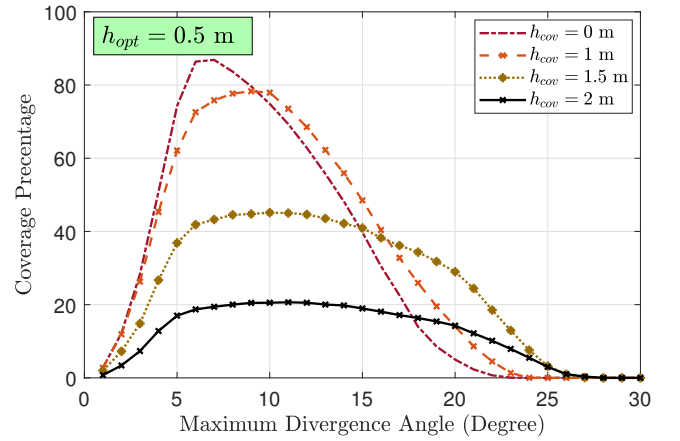


Fig. 7. Coverage as a function of $\theta_{\text{div},m}$ for $h_{\text{best}} = 0.5$ m and four different h_{cov} values.

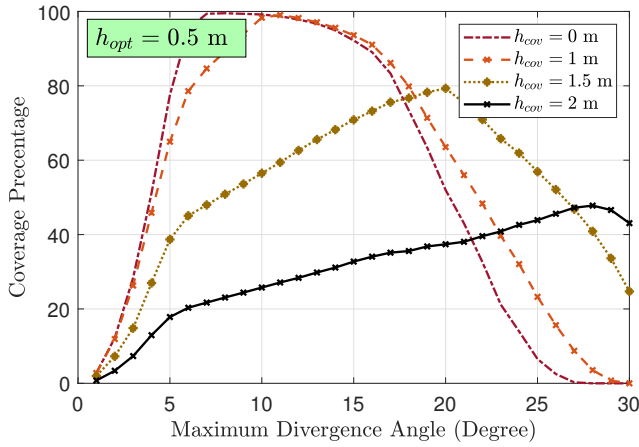


Fig. 8. Coverage as a function of $\theta_{div,m}$ for $h_{best} = 0.5$ m with two VCSEL laser links per user.

IV. CONCLUSION AND FUTURE REMARKS

This paper explored the potential of VCSEL-based indoor OWC systems to fulfill the critical performance requirements of 6G networks, including ultra-high data rates, energy efficiency, and flexible coverage. Through extensive simulations, we examined the influence of system parameters, including optimal height h_{best} , divergence angle θ_{div} , and the allocation of VCSEL beams per user, on system performance metrics like 3D coverage and network capacity.

Our findings reveal that optimizing h_{best} for maximum coverage at higher heights enhances 3D coverage performance but it might reduce network capacity by decreasing the number of available VCSELs to serve users at lower heights. Furthermore, by allocating multiple VCSEL beams to a single user and prioritizing coverage over spatial multiplexing, the system can achieve 100% coverage.

This study opens avenues for future research. One promising direction is the exploration of adaptive beam allocation algorithms that leverage machine learning to dynamically optimize h_{best} , θ_{div} , and beam allocation based on real-time user mobility and traffic patterns. Additionally, integrating VCSEL-based OWC with other wireless technologies, such as millimeter-wave and terahertz communication, could further enhance coverage flexibility and capacity in heterogeneous 6G networks. Finally, experimental validation of the proposed configurations in realistic indoor environments will be critical to bridge the gap between theoretical findings and practical implementations.

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REFERENCES

- [1] C. -X. Wang *et al.*, "On the Road to 6G: Visions, Requirements, Key Technologies, and Testbeds," *IEEE Commun. Surv. & Tutor.*, vol. 25, no. 2, pp. 905-974, Feb. 2023.
- [2] Krishnamoorthy, Aravindh, *et al.*, "Optical Wireless Communications: Enabling the Next Generation Network of Networks," *arXiv:2412.16798*, 2024.
- [3] M. D. Soltani *et al.*, "Terabit Indoor Laser-based Wireless Communications: LiFi 2.0 for 6G," *IEEE Wirel. Commun.*, vol. 30, no. 5, pp. 36-43, Oct. 2023.
- [4] HB Jeon *et al.*, "Free-Space Optical Communications for 6G Wireless Networks: Challenges, Opportunities, and Prototype Validation," *IEEE Commun. Mag.*, vol. 61, no. 4, pp. 116-121, Apr. 2023.
- [5] C. Chen *et al.*, "100 Gbps Indoor Access and 4.8 Gbps Outdoor Point-to-point LiFi Transmission Systems using Laserbased Light Sources," *J. Lightw. Technol.*, vol. 42, no. 12, pp. 4146-4157, May 2024.
- [6] K. Katsaros *et al.*, "AI-Native Multi-Access Future Networks—The REASON Architecture," *IEEE Access*, vol. 12, pp. 178586-178622, Nov. 2024.
- [7] M. T. Dabiri *et al.*, "Iterative signal detection under timing errors for optical wireless links with high mobility," *IEEE Trans. Veh. Technol.*, vol. 70, no. 11, pp. 11710-11720, Nov. 2021.
- [8] H. Kazemi *et al.*, "A Tb/s Indoor MIMO Optical Wireless Backhaul System Using VCSEL Arrays," *IEEE Trans. Commun.*, vol. 70, no. 6, pp. 3995-4012, Apr. 2022.
- [9] R. Michalzik, *VCSELs: Fundamentals, Technology and Applications of Vertical-Cavity Surface-Emitting Lasers*. Springer, 2012.
- [10] Y. Liu *et al.*, "High-Capacity Optical Wireless VCSEL Array Transmitter With Uniform Coverage," *SPIE Free-Space Laser Commun. XXXV*, vol. 12413, p. 124130J, Mar. 2023.
- [11] H. Kazemi *et al.*, "Multi-Beam Access Point Design for 6G Laser-Based Optical Wireless Networks: Eye Safety-Coverage Tradeoff," *IEEE Glob. Commun. Conf.*, Kuala Lumpur, Malaysia, pp. 5543-5548, Dec 2023.
- [12] H. Kazemi, *et al.*, "A Novel Terabit Grid-of-Beam Optical Wireless Multi-User Access Network With Beam Clustering," *arXiv:2404.04443*, Apr. 2024.
- [13] Z. Zeng, M. D. Soltani, M. Safari and H. Haas, "A VCSEL Array Transmission System With Novel Beam Activation Mechanisms," *IEEE Trans. Commun.*, vol. 70, no. 3, pp. 1886-1900, Mar. 2022.
- [14] M. Zohrabi, R. H. Cormack, and J. T. Gopinath, "Wide-angle Nonmechanical Beam Steering Using Liquid Lenses," *Opt. Exp.*, vol. 24, no. 21, pp. 23798-23809, Sep. 2016.
- [15] V. Mai and H. Kim, "Non-mechanical Beam Steering and Adaptive Beam Control Using Variable Focus Lenses for Free-space Optical Communications," *J. Lightw. Technol.*, vol. 39, no. 24, pp. 7600-7608, Dec. 2021.
- [16] H. Roscher, F. Rinaldi, and R. Michalzik, "Small-Pitch Flip-Chip-Bonded VCSEL Arrays Enabling Transmitter Redundancy and Monitoring in 2-D 10-Gbit/s Space-Parallel Fiber Transmission," *IEEE J. Sel. Top. Quantum Electron.*, vol. 13, no. 5, p. 1279-1289, Oct. 2007.
- [17] J. Baker *et al.*, "VCSEL Quick Fabrication for Assessment of Large Diameter Epitaxial Wafers," *IEEE Photon. J.*, vol. 14, no. 3, p. 1-10, June 2022.
- [18] Saleh BE, Teich MC., *Fundamentals of Photonics*. John Wiley & Sons, 2019.